CONJECTURE C, THE COLLAR FAMILY AND THE CENSUS OF TILE COUNTS, WITH COMPUTER EVIDENCE (COMPANION TO PAPER W)

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Abstract. A positive integer N is a *tile count* if some triangle can be cut into N pairwise congruent triangles (Problem 634 of `erdosproblems.com`). This is the companion of Paper W [14], which proves that the triangles $W_M(e,f)$ and $T_M(e,f)$ built from the tile (ef, f^2-e^2, f^2) are tiled once $Me \geq f$ and $(M-1)(f-e) \geq f$ (Theorem W); that theorem is used here, not proved. *The prime case.* The statement that no prime $N \equiv 3 \pmod 4$, $N > 3$, is a tile count is stated and machine-checked in the author's Lean development and was published in the release of 15 October 2026 (Zenodo 10.5281/zenodo.23400433); it has had internal audits only and no outside check, and the author makes no claim beyond that. Parallel work: Paliy (repository note, 29 September 2026) and Beeson–Roşu (arXiv:2607.23453, v2 16 October 2026, v3 21 October 2026). This paper collects what surrounds it. (1) A *collar family*: repeated one-layer steps give the tile counts $(f^2-1)(2f^2-1)g^2$ for $g \geq f+1$, and tails $465g^2$ ($g \geq 5$ in Lean; $g = 2,3,4$ by exact tilings), $966g^2$ and $2485g^2$ ($g \geq 2$). (2) Twenty-six further tile counts by macro-piece layouts found by a template search, each with a Lean proof, and a twenty-seventh by square closure. (3) *Conjecture C*, a candidate description of all tile counts as a union of explicit families: its existence half is proved on the ranges known, its exclusion half is open. (4) A census of the non-squarefree $N \leq 10^6$, by exact computation. (5) Computer evidence, not a proof, on one open family. (6) The *thick regime*: Lean and hand no-tiling theorems for $W_M(e,f)$ when $M < f$, $Me \geq f$ and $M(f-e) \leq e+2$, with the census effect, and a statement of what remains open. Every statement carries one of five labels (Lean, hand proof, exact computation, claimed by Paliy, conjecture); a list of corrections of earlier statements and a list of what is not proved are included. The prime case is not treated.

1. Introduction

1.1. **The problem and what this paper adds.** A triangle Ω (the *target*) is *tiled by N copies of a triangle* tile (the *tile*) if Ω is the union of N closed triangles with pairwise disjoint interiors, each congruent to tile; mirror images are allowed and the pieces need not meet edge to edge. N is a *tile count* if some triangle is tiled by N copies of some triangle (in the Lean statements, `TileCount N` says exactly this). The squares, the numbers x^2+y^2, $3k^2$ and $6k^2$ are tile counts for classical reasons [18, 17]. Laczkovich classified the pairs (target, tile) up to the arithmetic of their angles [10, 11]; Beeson, with Roşu, Zhang and Laczkovich, worked through the classes [1, 6, 7, 2, 19, 9]. Paliy published constructions and a proof of the prime case [12, 13]. The earlier versions of this paper [16, 15] contain the exclusion side (the primes $p \equiv 3 \pmod 4$, $p > 3$, are not tile counts, a theorem of the earlier release that has been audited only inside the project, see Section 6; every $N \leq 200$ is decided; a census of the non-squarefree $N \leq 10^6$). This paper (version 4.0) is mainly about the other side, *which counts are tile counts*, and is organised around one family of tiles, the one that carries every open question of the census; it also adds exclusion results (Sections 7 and 8).

What we take from the earlier versions. Four facts are used as black boxes, each a Lean theorem of the earlier release [15, 16] that is not reproved here: for the tile $\mathsf{tile}(e,f)$ of Section 2.1

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and the target $W_M(e,f)$, $T_M(e,f)$ of Section 2.2, (i) $W_M(e,f)$ is not tiled if $Me < f$ (the below-threshold corollary of the earlier release, part (a)), and $T_M(e,f)$ is not tiled if $2Me < f$ (same corollary, part (b)); (ii) $W_M(e,f)$ is not tiled if $M(f-e) \le e$ (thick-regime corollary of the earlier release, part (a)); (iii) for $e = 1$ and for $e = f-1$ the scale $M = f$ is the threshold for W_M (the same two corollaries, part (b)); (iv) W_M and T_M are tiled for every $M \ge f$ (Paliy, Beeson; formalized there).

Standing of the statements. Every statement carries one of five labels:

- [PROVED IN LEAN] a theorem of the accompanying Lean 4 library, checked by the kernel with the standard axioms only (`propext`, `Classical.choice`, `Quot.sound`), no `sorry`. The Lean statement has been compared with the mathematical one *within this work only*; a reading by a person outside the project has not been made.
- [PROVED BY HAND] a written proof, audited inside the project, not formalised.
- [COMPUTED, EXACT] an exact finite computation, typically an explicit tiling whose pieces are listed in exact arithmetic and verified by an independent exact checker. For an explicit tiling this is a proof of existence, but not a Lean proof.
- [CLAIMED BY PALIY] a public claim of D. Paliy that we have not re-proved (in some cases we have checked an example).
- [CONJECTURE] unproved.

Where a result has two labels (say, a hand proof and a Lean proof) both are given.

1.2. **Results at a glance.** In detail: (1) **Theorem W** (Paper W [14], Theorem 2.1; Lean and hand proof there): for $2 \le e \le f-2$, $W_M(e,f)$ is tiled by $M^2(2f^2-e^2)$ tiles, and $T_M(e,f)$ by $M^2(3f^2-e^2)$ tiles, as soon as $Me \ge f$ and $(M-1)(f-e) \ge f$; stated here in Section 3 and used, not proved. (2) *Collar family* (Lean): a collar construction for the isosceles target with base angles α gives, by repeated one-layer steps, the counts $(f^2-1)(2f^2-1)g^2$ for all $g \ge f+1$ ($f \ge 2$), the tails $465\,g^2$ ($g \ge 5$), $966\,g^2$ ($g \ge 17$) and $2485\,g^2$ ($g \ge 5$); together with the counts of (3) at small g this gives $966g^2$ and $2485g^2$ for all $g \ge 2$ (Lean). New: $465g^2$ for $g = 2,3,4$ (1860, 4185, 7440) are tile counts of the tile $(4,15,16)$ by exact-verified tilings (not yet Lean), so $465g^2$ is a tile count for every $g \ge 2$; the core 465 ($g = 1$) stays open. The up-set in g was announced earlier by Paliy. (3) *Macro pieces*: twenty-six tile counts have individual Lean proofs, by macro-piece layouts found by a template search and verified exactly, and a twenty-seventh (78 246) follows by square closure; nineteen of them are values of $966g^2$ and $2485g^2$ at small g (which complete the tails of (2); some also follow from others by Proposition 2.2), and seven are not (375, 540, 735, 825, 630, 1120, 1750). One hundred tilings are shipped as exact data. (4) **Conjecture C** (conjecture) states the set of tile counts as a union of explicit families; its existence half is a Lean theorem for the ranges proved so far (a collection narrower than the full table of families), and its exclusion half is open. (5) *Census* (exact computation, partly formal; version 3.0 figures, updated in Section 8 to 200 282 excluded and 62 421 undecided): for the non-squarefree $N \le 10^6$, 199 364 values are excluded by Lean theorems (the assignment is made by a program; the theorems have been audited only inside the project), 129 108 are tile counts, 63 602 are undecided; counting the hand-proved Theorem R2 class as well, 199 441 / 129 108 / 63 525; about 770 squarefree values stay open. The three values 1860, 4185, 7440 above were found after the census was run and are not in these numbers. (6) **The thick regime** (Section 7): for coprime $2 \le e < f$, $M < f$, $Me \ge f$: $W_M(e,f)$ has no tiling if $M(f-e) \le e+1$ (Lean; hand proof audited inside the project; 124 census values; formally in the exclusion chain: 412, 270), and, as a hand proof only, if $M(f-e) = e+2$ (77 more values).

The table gives the same results with their labels and places.

result	statement in one line	label and place
Theorem W (Paper W; §3 here)	$W_M(e,f)$ and $T_M(e,f)$ tile once $Me \geq f$ and $(M-1)(f-e) \geq f$ $(2 \leq e \leq f-2)$; Lean: the tiling for W_M, the count for T_M	W half [PROVED IN LEAN] [PROVED BY HAND]; count of T_M [PROVED IN LEAN]; tiling of T_M via an imported glue lemma; proved in Paper W
Sharpness (Paper W, Cor. 2.4)	for $2 \leq e < f/2$: $W_M(e,f)$ tiles iff $Me \geq f$	"if" [PROVED IN LEAN]; "only if" imported from the earlier release
Collar family (§4)	$(f^2-1)(2f^2-1)g^2$, $g \geq f+1$; tails $465g^2$ ($g \geq 5$ [PROVED IN LEAN]; $g = 2,3,4$ [COMPUTED, EXACT] exact tilings only), $966g^2$ and $2485g^2$ ($g \geq 2$, using the macro counts at small g)	[PROVED IN LEAN]; collar step with arbitrary height [PROVED BY HAND]
Macro-piece counts (§5)	27 further tile counts (e.g. 3864, 9940, 62125, 279174)	[PROVED IN LEAN], [COMPUTED, EXACT] data
Conjecture C (App. B)	the set of tile counts is $\bigcup_k k^2 \cdot \{\text{families}\}$	existence half [PROVED IN LEAN] on a narrower collection of proved ranges (App. B); exclusion half [CONJECTURE]
Census (App. C)	version 4.0: 200 282 excluded / 62 421 undecided (Section 8); version 3.0: 199 364 / 129 108 / 63 602 (non-squarefree $N \leq 10^6$; with the hand-proved class R2: 199 441 / 129 108 / 63 525)	[COMPUTED, EXACT] with label columns
Thick regime (§7)	$W_M(e,f)$ has no tiling for $e \geq 2$, $M < f$, $Me \geq f$, $M(f-e) \leq e+1$; and for $M(f-e) = e+2$	first: [PROVED IN LEAN] (hand proof audited inside the project); second: [PROVED BY HAND] only (arithmetic half in Lean)
W^{3+} evidence (App. D)	a set of candidate boundary words of a tiling of an isosceles target, defined in the appendix; for $(4,7)$, $(5,8)$ at $g=3$ no tiling was found for any of these words (a verdict conditional on hand-proved filters)	[COMPUTED, EXACT] (not formal)

1.3. **Reading guide.** The body of the paper (Sections 2–9) contains the new constructions: Section 2 defines the tile, the targets and the two tools used throughout (ladder subdivision, square closure); Section 3 restates Theorem W of Paper W and what it does to this paper; Sections 4 and 5 give the collar family and the macro-piece counts; Section 6 lists what is not proved and Section 9 gives credits with dates. The appendices hold the material that is needed for checking or for context: Appendix A says where the proof of Theorem W is (in Paper W); Appendix B states Conjecture C, a candidate description of all tile counts, and what is proved about it; Appendix C is the census; Appendix D is computer evidence on one open family; Appendix E lists corrections of earlier statements; Appendices F–H give the data, an index of the Lean statements and the reproduction commands. Three levels: ten minutes (this

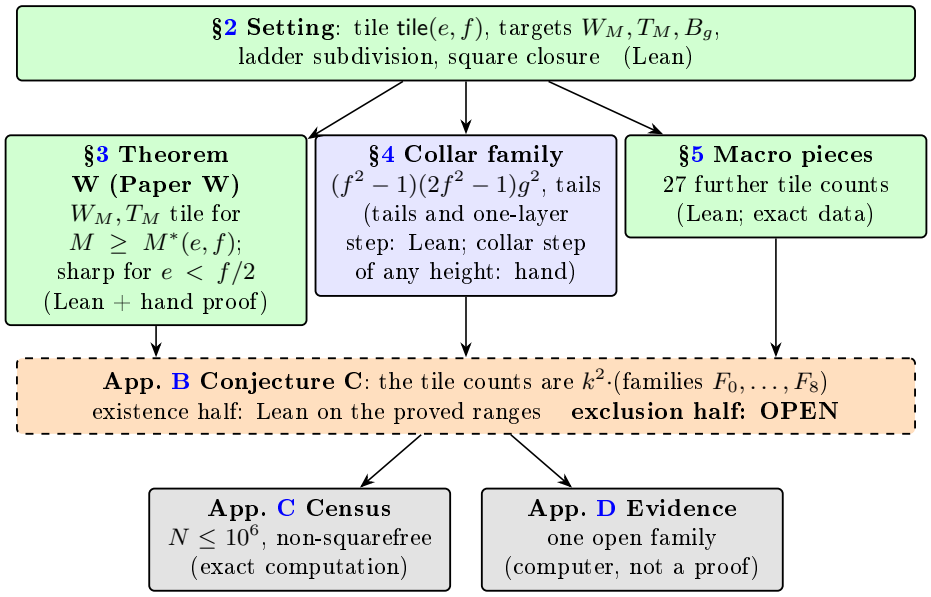


FIGURE 1. Map of the paper. Green: proved in Lean (Theorem W has an additional hand proof); blue: hand proof (the collar step of arbitrary height; only height one and the tails are Lean); grey: exact or computer evidence only; orange dashed: conjecture. This figure shows the construction side only; the exclusion results are in Sections 7 and 8.

introduction and Figure 1); an hour (Sections 2–3 and the part of Section 4 you care about); an afternoon (the body and Appendix B). Everything else is supplementary and can be skipped.

1.4. **Credit and the use of an AI system.** The constructions in Sections 3–5 build on those of D. Paliy (the W/β framework, caps and collars, the α-isosceles up-set, seeds), M. Beeson and G. Roşu (the targets W, T and the methods), B. Herdt [4], M. Laczkovich, Y. X. Zhang and J. P. Harries [5]; Section 9 says exactly which statement is whose, with dates. These results were obtained with the help of an AI system (Claude, Anthropic); the questions and the directions were the author's, and the proofs have been checked in Lean or by hand as labelled; the provenance table is in the repository README.

2. SETTING

2.1. **The tile.** Throughout, $e < f$ are coprime positive integers and

$$a = ef, \qquad b = f^2 - e^2, \qquad c = f^2, \qquad \kappa = b + c = 2f^2 - e^2.$$

The *tile* $\mathsf{tile}(e, f)$ is the triangle with sides a, b, c; α, β, γ are its angles opposite a, b, c. Heron's formula gives $16\,\mathrm{Area}^2 = e^2 b\,(\kappa^2 - a^2) = e^2(f^2 - e^2)(ef + 2f^2 - e^2)(2f^2 - e^2 - ef)$ (for the tile $(2, 3, 4)$, $e = 1, f = 2$: 135), and one checks $c^2 = a^2 + bc$ and

$$3\alpha + 2\beta = \pi, \qquad \gamma = 2\alpha + \beta, \qquad \cos\alpha = \frac{\kappa}{2c}.$$

(In the classification of the earlier paper these are the tiles of "type (T)", the family on which its prime case rests; nothing below depends on that.) That (a, b, c) is a triangle follows from $a + b - c = e(f - e) > 0$. The angle α/π is irrational; this is not used below. (Figure 2.) Small examples: $(e, f) = (1, 2)$ gives the tile $(2, 3, 4)$; $(2, 3)$ gives $(6, 5, 9)$; $(2, 5)$ gives $(10, 21, 25)$; $(1, 4)$ gives $(4, 15, 16)$.

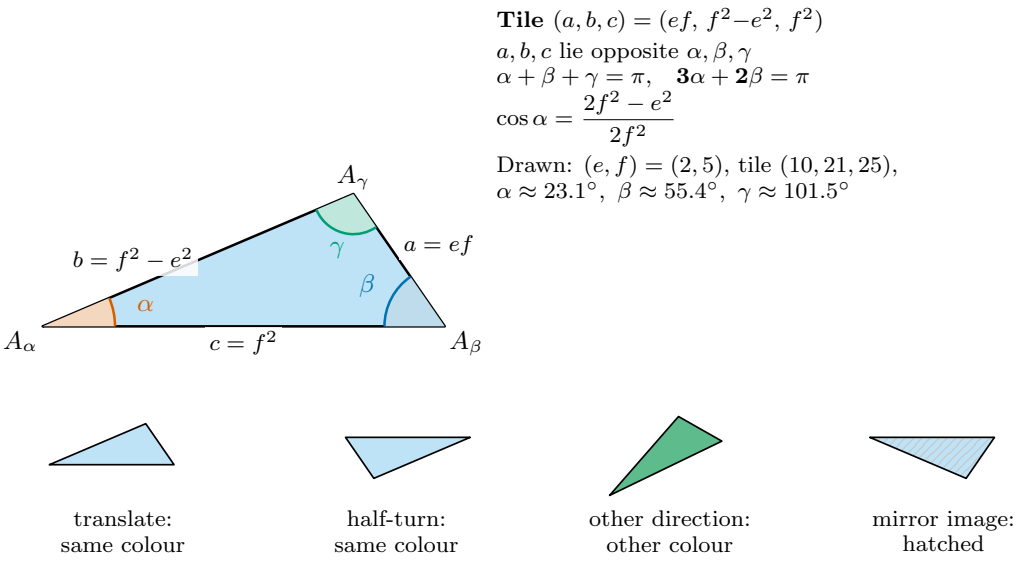


FIGURE 2. The tile $\mathsf{tile}(e,f)$ with sides $a=ef$, $b=f^2-e^2$, $c=f^2$ and angles α (opposite a), β (opposite b), $\gamma=2\alpha+\beta$ (opposite c), drawn for $(e,f)=(2,5)$, the tile $(10,21,25)$. The relation $3\alpha+2\beta=\pi$ is what makes the angles of the tile combine into the angles of the targets below. In all tiling figures that follow, tiles of the same orientation (up to a half-turn) share a colour, mirror images are hatched, and heavier lines are the boundaries between regions of one colour.

2.2. **The targets.** Let $M\geq 1$ be an integer. The three families of targets that occur are listed in the table; the last column is the number of tiles that fit by area, which is the only count a tiling of the target can have.

target	sides	angles	area in tiles
$T_M(e,f)$	$Mf^3,\ Mf^3,\ Me(3f^2-e^2)$	isosceles; apex 3α, base angles β	$M^2(3f^2-e^2)$
$W_M(e,f)$	$Mfb,\ Me\kappa,\ Mf^3$	angles $2\alpha,\ \beta,\ \alpha+\beta$ (opposite $Me\kappa,\ Mfb,\ Mf^3$)	$M^2(2f^2-e^2)$
$B_g(e,f)$	$gbc,\ gbc,\ gb\kappa$	isosceles; base angles α, apex $\alpha+2\beta$	$b\kappa\, g^2$

(For W_M, the side opposite the angle 2α is $Me\kappa$, opposite β is Mfb, and opposite $\alpha+\beta$ is Mf^3.) T_M and W_M are the targets of Beeson–Roşu and Paliy ("base-β" targets); B_g is the "base-α" target (Paliy's α-isosceles). We call a target *tiled* if it is tiled by copies of the tile $\mathsf{tile}(e,f)$ of the same (e,f). $W_M(e,f)$ and $T_M(e,f)$ are related by gluing: a tile at scale Mf glued to the side Mfb of W_M gives T_M (Paper W, Proposition 2.3); all three are drawn in Figure 3.

2.3. **Two tools.**

Proposition 2.1 (Ladder subdivision; [PROVED IN LEAN]). *Let $s\geq 1$ be an integer. A triangle similar to* tile *with sides sa, sb, sc is tiled by s^2 copies of* tile*. A parallelogram with sides sa and uc enclosing the angle β is tiled by $2su$ copies of* tile *(each unit cell is a parallelogram with sides a, c enclosing β; the diagonal joining the far ends of those two sides has length b and cuts the cell into two tiles).*

Lean: `Subdivision.ladderDissection`, `Ladder.ladder`; the earlier paper calls this its ladder theorem. A *trapezoid* below means a triangle of scale s_1 minus a corner triangle of scale $s_2<s_1$ of the same shape (cut off by a line parallel to the opposite side); it contains $s_1^2-s_2^2$ tiles.

TABLE 1. Symbols. This table is shared word for word with Paper W [14].

symbol	meaning ([W] = Paper W; [C] = the companion paper)
used in both papers	
$e < f$	coprime positive integers defining the tile; $d = f - e$
a, b, c, κ	$a = ef$, $b = f^2 - e^2$, $c = f^2$, $\kappa = b + c = 2f^2 - e^2$: the sides of the tile $\mathsf{tile}(e, f)$ and the sum $b + c$
α, β, γ	angles of the tile opposite a, b, c; $3\alpha + 2\beta = \pi$, $\gamma = 2\alpha + \beta$
scale s	a triangle with sides sa, sb, sc has *scale* s and is tiled by s^2 tiles (ladder subdivision, [W])
W_M, T_M	the two targets of [W]; M is the *scale* of the target; W_M is tiled by $M^2(2f^2 - e^2)$ tiles, T_M by $M^2(3f^2 - e^2)$
$M^*(e, f)$	threshold scale $\max(\lceil f/e \rceil, \lceil f/(f-e) \rceil + 1)$ of Theorem W; (H1), (H2) are $Me \geq f$ and $(M-1)(f-e) \geq f$, together equivalent to $M \geq M^*(e, f)$
used in [W] only	
n, ρ, δ	$n = \lfloor eM/f \rfloor$, $\rho = eM - fn$, $\delta = 1$ if $\rho = 0$ and 0 otherwise
t, r, q	the three free integers of the six-piece layout; canonical choice $t = n - \delta$, $r = n + 1$, $q = n$
(ξ, y), (u, w)	skew frame with origin P ($OP = OB$) and lattice coordinates $u = (f\xi + ey)/efb$, $w = (e\xi + fy)/efb$; $\ell_k = \{u = k\}$ (Section 2.3 of [W])
v, s_1, s_2, x, u_1, u_2	the integer scales of the pieces, $v = fM - er$, $s_1 = f(r - t)$, $s_2 = fr - eM$, $x = ft - eq$, $u_1 = fq - et$, $u_2 = fq - er$
(K)	the four inequalities $ft < eM < fr$, $er < fq$, $eq < ft$
$\pi_0, \ldots, \pi_5$	the six pieces of the layout; $O, A, B, P, P_2, P_3, Q, R, B', C', V_2, X_4, U_1, U_2$ its points
used in [C] only	
B_g	the isosceles target with base angles α; g is its scale; $b\kappa g^2$ tiles
row (e, f)	the tile $\mathsf{tile}(e, f)$; $G(e, f)$ is the set of scales g for which $B_g(e, f)$ is tiled
$F_0, \ldots, F_8$	the families of Conjecture C

Proposition 2.2 (Square closure; [PROVED IN LEAN]). *If N is a tile count then so is k^2N for every integer $k \geq 1$.*

Lean: `TileCountFamilies.tileCount_scale`. (Enlarge the tiling of Ω by the factor k: this gives a tiling of a triangle $k\Omega$ by N copies of the tile $k\mathsf{tile}$, with all sides multiplied by k. Each copy of $k\mathsf{tile}$ is a triangle of scale k (sides ka, kb, kc), so by Proposition 2.1 it is tiled by k^2 copies of the original tile tile itself. In total $k\Omega$ is tiled by k^2N copies of tile.)

Remark 2.3. In words: scaling the whole figure by k multiplies the number of tiles by k^2. Because of Proposition 2.2 a tile count is naturally written $N = k^2n$, and thresholds below are thresholds in the *scale of the target relative to the tile*, not in k.

3. THEOREM W (OF PAPER W) AND WHAT IT DOES HERE

Everything in this section is proved in Paper W [14]; the statements are restated so that the rest of this paper can be read alone. Notation: $M^*(e, f) = \max(\lceil f/e \rceil, \lceil f/(f-e) \rceil + 1)$ and $d = f - e$.

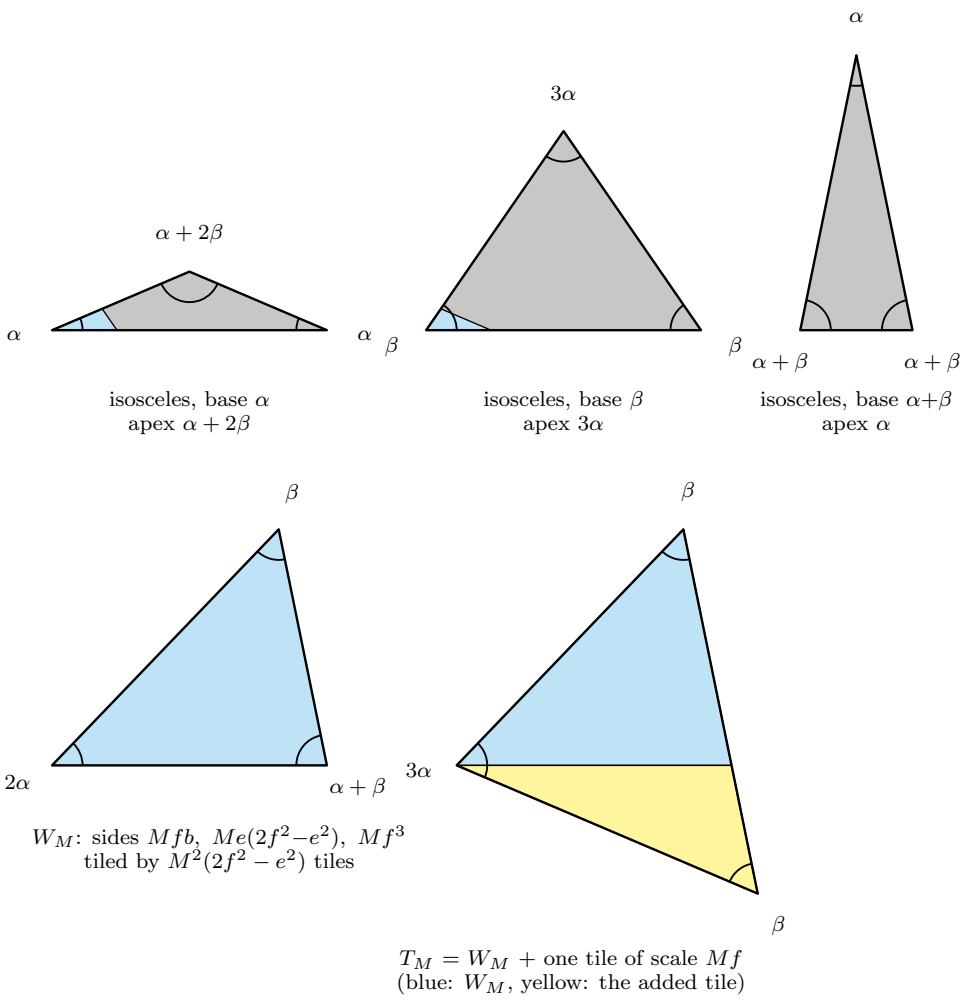


FIGURE 3. The targets for $(e, f) = (2, 5)$, $M = 1$ proportions. Top: three isosceles shapes with base angles α, β and $\alpha+\beta$, with one tile placed at a base corner; the first is B_g, the second is T_M in outline, and the third (base angles $\alpha+\beta$, apex α) is the θ-isosceles target of the family F_8 of Appendix B, not used before that section. Bottom left: W_M (angles 2α, $\alpha+\beta$, β), tiled by $M^2(2f^2-e^2)$ tiles. Bottom right: T_M, obtained from W_M by adding one tile of scale Mf along the side Mfb; it is the isosceles triangle with base angles β and apex 3α, tiled by $M^2(3f^2 - e^2)$ tiles. Angles are built from α, β by $3\alpha + 2\beta = \pi$.

Theorem 3.1 (Theorem W; Paper W, Theorem 2.1; W half [PROVED IN LEAN] and [PROVED BY HAND], count of T_M [PROVED IN LEAN], tiling of T_M via a glue lemma of the earlier release)**.** *Let e, f be coprime with $2 \le e \le f - 2$ and $M \ge 1$ with (H1) $Me \ge f$ and (H2) $(M-1)(f-e) \ge f$. Then $W_M(e, f)$ is tiled by $M^2(2f^2-e^2)$ copies of* tile(e, f)*, and $T_M(e, f)$ by $M^2(3f^2-e^2)$ copies.*

Corollary 3.2 (Paper W, Corollary 2.4; "if" [PROVED IN LEAN], "only if" imported from the earlier release)**.** *For coprime $2 \le e < f/2$, $W_M(e, f)$ is tiled if and only if $Me \ge f$.*

Corollary 3.3 (Paper W, worked example in Section 5)**.** *$W_4(3, 5)$, with sides $320, 492, 500$, is tiled by 656 copies of $(15, 16, 25)$. The earlier paper listed this case as open.*

Role here. Theorem W supplies the family F_1 of Conjecture C and 26 138 of the 26 312 new values of the census that are Lean theorems (Appendix C; the rest of the 26 312 are the collar family, tails, macro counts, the thirteen witnesses and k^2-multiples); the proof (a layout of six pieces, with a table of every sign condition and the exact statement of what is Lean, hand, computed or imported) is Paper W. The corrections that Paper W makes to earlier expectations are listed in Appendix E. What stays open for W_M, T_M (the "no" side for $e > f/2$, $e = 1$ and $e = f - 1$ for T_M, the T half in general) is stated in Paper W, Section 6, and summarised in Section 6. The glue that gives T_M from W_M (one tile of scale Mf on the side Mfb) is Paper W, Proposition 2.3 (Section 2.2).

4. The collar family and the base-α rows

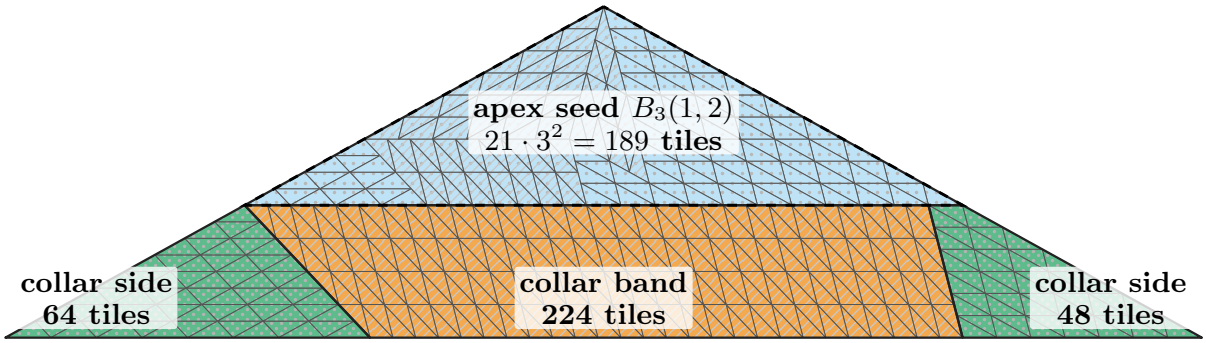
This section is about the third target of Section 2.2, the isosceles triangle $B_g(e, f)$ with base angles α, legs gbc, base $gb\kappa$, and $b\kappa g^2$ tiles. Write $N_0(e, f) = b\kappa = (f^2 - e^2)(2f^2 - e^2)$, so that a tiling of $B_g(e, f)$ has $N_0(e, f)g^2$ tiles; *the row* (e, f) is the set of g for which $B_g(e, f)$ is tiled. The first values: $N_0(1, 2) = 21$, $N_0(2, 3) = 70$, $N_0(1, 4) = 465$, $N_0(2, 5) = 966$, $N_0(1, 6) = 2485$. The row is non-empty in every case examined, and it is closed under addition (Paliy, [CLAIMED BY PALIY]; his construction), so it is, if his claim holds, a numerical semigroup; what is open is its smallest element and its gaps.

4.1. **A priority statement.** D. Paliy stated on 29 September 2026 [13] that, for each tile, the targets B_K (which have $b\kappa K^2$ tiles) are tiled for every K beyond an explicit threshold K_0, so that the tile counts $b\kappa K^2$ are realised (the "α up-set"); the seeds 84 and 189 of the row $(1, 2)$ are in Beeson's work (the tilings of 84 and 189 in his Figures 28 and 29; 189 is due to Harries). **The up-set and the counts are Paliy's.** What is ours in this section is an explicit construction of the layers (the *collar*), a parametric seed for the whole family $e = 1$, the explicit thresholds that follow, and their Lean proofs; the construction is independent of his, and the thresholds of Theorem 4.3 for the rows $(1, 4)$ and $(1, 6)$ are lower than the ones obtained from his formula as we read it, and for $(2, 5)$ our tail coincides with his before the macro counts of Section 5 fill the smaller g (we have not asked him about these readings).

4.2. **The collar step.**

Theorem 4.1 (Collar step; [PROVED BY HAND], with a [PROVED IN LEAN] one-layer case)**.** *Let $e < f$ be coprime, $a = ef$, $b = f^2 - e^2$, $\kappa = 2f^2 - e^2$, and suppose $B_{g_0}(e, f)$ is tiled. Choose $\sigma \in [0, a)$ with $b\sigma \equiv b\kappa g_0 \pmod a$ (it exists since $\gcd(a, b) = 1$), put $K = (b\kappa g_0 - b\sigma)/a$ and $h_{\max} = \lfloor K/a \rfloor$. Then for every integer h with $1 \le h \le h_{\max}$ the triangle $B_{g_0+h}(e, f)$ is tiled.*

Construction. B_{g_0+h} is a homothetic copy of B_{g_0} at the apex plus a trapezoidal strip along the base; the strip is cut into three pieces by two lines, each tiled by Proposition 2.1: a triangle of scale ch, the difference of two triangles $\Delta(K) \setminus \Delta(K - ah)$, and the difference $\Delta(\sigma + bh) \setminus \Delta(\sigma)$. The identity that makes the cut edges coincide is $c^2h + a(K - ah) = b\kappa(g_0 + h) - b(\sigma + bh)$, which reduces to $c^2 - a^2 = bc$; the range $h \le K/a$ is exactly the condition $K - ah \ge 0$. *Status.* The hand proof was audited independently (verdict: sound; checked exactly on 17 853 instances, all coprime $e < f < 40$ with several (g_0, h) each, and no failure). [PROVED IN LEAN] The case $h = 1$, for every row with $ef \le f^2 - e^2$, is Lean `Draft.CollarFam.layer_step` (real-level core `Draft.CollarLayer.layer_core`); arbitrary h is hand. The corollary "g_0 in the row implies every $g \ge g_0$ is in the row" needs $h_{\max}(g_0) \ge g_0 - 1$ together with $h_{\max}(x) \ge 1$ for $x \ge 2g_0 - 1$ (both conditions are needed; see Appendix E); it holds for the rows below. Figure 4 shows a collar around a seed.



$B_5(1,2) = B_3(1,2) \cup$ collar: $21 \cdot 5^2 = 525$ tiles of $(2,3,4)$ in a triangle $(105, 60, 60)$

FIGURE 4. The collar construction for the tile $(2,3,4)$ (row $(1,2)$): the triangle $B_5(1,2)$ with $525 = 21 \cdot 5^2$ tiles is an apex seed $B_3(1,2)$ (189 tiles, the top triangle) together with a collar of 336 tiles (a band along the base and two side pieces). Colours are orientation classes. This is the picture of what the tiling contains; the step of Theorem 4.1 for general g_0, h cuts the strip into three ladder pieces as described in the text.

4.3. The seed for $e = 1$.

Theorem 4.2 (Seed and up-set for $e = 1$; [PROVED IN LEAN]). *For every $f \geq 2$ and every $g \geq f+1$, $(f^2-1)(2f^2-1)\,g^2$ is a tile count; the tiling is of $B_g(1,f)$ by copies of (f, f^2-1, f^2).*

Lean: `Draft.CollarUp.tileCount_e1_up`. The seed $g = f+1$ is a six-piece layout whose pieces are polynomial in f (generated and checked symbolically; module `CollarSeed`, 125 lemmas), and the layer step then gives all $g \geq f+1$. For $f = 2, 3, 4, 6$ the seed counts are 189, 2176, 11 625, 121 765; the first is the 189 of Beeson and Harries, the third and fourth were found by a template search in release v1.16.

Why it matters. The sporadic-looking counts 189, 11 625, 121 765 of earlier releases are the first members of one family, and the count $(f^2-1)(2f^2-1)g^2$ is a tile count for all $f \geq 2$ at scales that no longer depend on a search.

4.4. Tails for three rows.

Theorem 4.3 (Tails; [PROVED IN LEAN]). *(a) $465\,g^2$ is a tile count for every $g \geq 5$ (row $(1,4)$, tile $(4,15,16)$). (b) $2485\,g^2$ is a tile count for every $g \geq 2$ (row $(1,6)$, tile $(6,35,36)$). (c) $966\,g^2$ is a tile count for every $g \geq 2$ (row $(2,5)$, tile $(10,21,25)$).*

Lean: (a) `Draft.CollarFam.tileCount_465_sq`, from the seed 11 625 and the layer step. (b) The tail $g \geq 5$ is `Draft.CollarFam5.tileCount_2485_sq5`; $g = 2, 3$ are 9940 and 22 365 (Section 5), and $g = 4$ is $4 \cdot 9940$ by Proposition 2.2. (c) The tail $g \geq 17$ is `Draft.CollarFam17.tileCount_966_sq17`; $g = 2, \ldots, 8$ and $10, \ldots, 18$ are the macro-piece counts $3864, \ldots, 312\,984$ of Section 5 (sixteen values), and $g = 9$ is $9 \cdot 8694 = 78\,246$ (`Draft.AllMacro.tileCount_78246`). The squarefree cores 465, 966 and 2485 themselves ($g = 1$) are not tile counts known to us; they are among the open squarefree values.

Remark 4.4. In our working notes we had recorded the pattern "the smallest scale of a row is $f+1$ for even f". (b) refutes it (the row $(1,6)$, with $f = 6$ even, contains $g = 2$, as does the row $(1,2)$), and (c) shows that the row $(2,5)$ contains $g = 2$ as well. For $(1,4)$ the values $g = 2, 3, 4$ (1860, 4185, 7440) are tile counts by exact-verified tilings of the base-α isosceles targets $B_g(1,4)$ (sides $240g, 240g, 465g$) by the tile $(4,15,16)$, found after the Lean work of this release (data files

`tiling_1860_s3_14_run_4_15_16_g2` and the two others; the exact verifier says VERIFIED; [COMPUTED, EXACT], not yet Lean; the Lean certificates and the use of $g_0 = 2$ as a seed of the layer step are planned). Hence $465g^2$ is a tile count for every $g \geq 2$ ($g \geq 5$ in Lean, $g = 2, 3, 4$ exact only). The core 465 ($g = 1$) is open; an exact search of bounded depth found no closure. The up-set in g and the threshold $K_0 \geq \lceil B/v \rceil$ are Paliy's; they are the reason $g \geq 5$ is a Lean theorem for $(1, 4)$, and the new exact seeds lower the threshold in practice. Paliy's threshold, read from his formula, is $K_0 = 17$ for $(2, 5)$ (equal to our tail in (c) before the macro pieces fill $g = 2, \ldots, 16$), 6 for $(1, 4)$ and 8 for $(1, 6)$ (we did not check these values with him).

5. Macro-piece tile counts

5.1. **The idea.** In the six-piece layout of Paper W every piece is tiled by the ladder subdivision, that is, by a tiling of a *triangle similar to the tile*. A *macro piece* is a piece that is a triangle similar, with integer ratio k, to the *target* of a tiling already known: such a piece is tiled by $k^2 n$ tiles if the known tiling has n tiles (Lean: `Draft.MacroPiece.macroTFam_gen`, which transports a known dissection by the ladder theorem and congruence of triangles with equal sides). Macro pieces were used for a long time by hand (Laczkovich's macro-dissections, Herdt's 1125; Section 6 of the version-2.2 paper), and they enlarge the set of layouts for which the area equation has integer solutions. The layouts are found by a computer search over the combinatorial type of the layout and the integer scales of its pieces, with the equations solved exactly; a layout is then expanded to the full list of tiles, verified by an exact checker, and finally turned into a Lean module, whose proof is the same area-and-containment argument as in Theorem W with the numbers of that layout. Figure 5 shows one such composition. The search itself is not part of this release; what is shipped are the results: the exact tilings (ninety-seven files, Appendix F) and the Lean theorems.

5.2. **The counts.**

Theorem 5.1 (Macro-piece tile counts; [PROVED IN LEAN], [COMPUTED, EXACT] data)**.** *Each of the following is a tile count, as the statement* `TileCount` N *of a Lean theorem:*

values N	*form*	*tile*	*what is tiled*
375, 540, 735	$15m^2$, $m = 5, 6, 7$	$(3, 5, 7)$	*equilateral triangles of side* $75, 90, 105$
825	$33 \cdot 5^2$	$(3, 5, 7)$	*isosceles, sides* $105, 105, 165$
630, 1120, 1750	$70\,g^2$, $g = 3, 4, 5$	$(6, 5, 9)$, *row* $(2, 3)$	$B_g(2, 3)$
3864, 8694, 15 456, 24 150, 34 776, 47 334, 61 824	$966\,g^2$, $g = 2, \ldots, 8$	$(10, 21, 25)$, *row* $(2, 5)$	$B_g(2, 5)$
96 600, 116 886, 139 104, 163 254, 189 336, 217 350, 247 296, 279 174, 312 984	$966\,g^2$, $g = 10, \ldots, 18$	$(10, 21, 25)$	$B_g(2, 5)$
9940, 22 365, 62 125	$2485\,g^2$, $g = 2, 3, 5$	$(6, 35, 36)$, *row* $(1, 6)$	$B_g(1, 6)$

That is 26 *values, and a twenty-seventh,* $78\,246 = 9 \cdot 8694$*, by Proposition 2.2.*

Lean: one module `Draft/TM`N (or `Draft/T62125`) per value, with `theorem tileCount_`N `: TileCount` N; aggregate `Draft.AllMacro`. Each module fixes the layout, proves the shapes and the containment by the area-and-sign method, and does not use a tile list; the standard axioms only. The exact tilings of all of them are in `data/witnesses/v3_exact` (for $N = 375$,

`tiling_375_macro_5_3_7_g1`); they have been verified by an exact checker independently of the Lean proofs.

Why it matters. Together with Theorem 4.3 these counts show that in two rows, $(2, 5)$ and $(1, 6)$, *every* scale $g \geq 2$ is realised; the squarefree cores of those rows are the only open members. The $15m^2$ values for $m = 5, 6, 7$ are the seeds that Paliy used ([CLAIMED BY PALIY]) for his claim that $15m^2$ is a tile count for every $m \geq 4$.

For the equilateral counts $15m^2$: $m = 4$ (240) was already a Lean theorem, and so was every m in the semigroup generated by 4 and 9 (earlier release). The Lean theorem `Draft.EqSemigroup.eqD_step` says that equilateral tilings of sides $15m_1$ and $15m_2$ by copies of $(3, 5, 7)$ give one of side $15(m_1 + m_2)$; the seeds $4, 5, 6, 7$ would therefore give every $m \geq 4$ (this is Paliy's claim, [CLAIMED BY PALIY]). The new Lean theorems for $m = 5, 6, 7$ state only `TileCount` and are not yet instances of the equilateral predicate used by `eqD_step`, so "$15m^2$ for all $m \geq 4$" is *not* a Lean theorem here; the exact tilings (sides $75, 90, 105$) are verified, and $15m^2$ for the individual $m = 5, 6, 7$ is Lean.

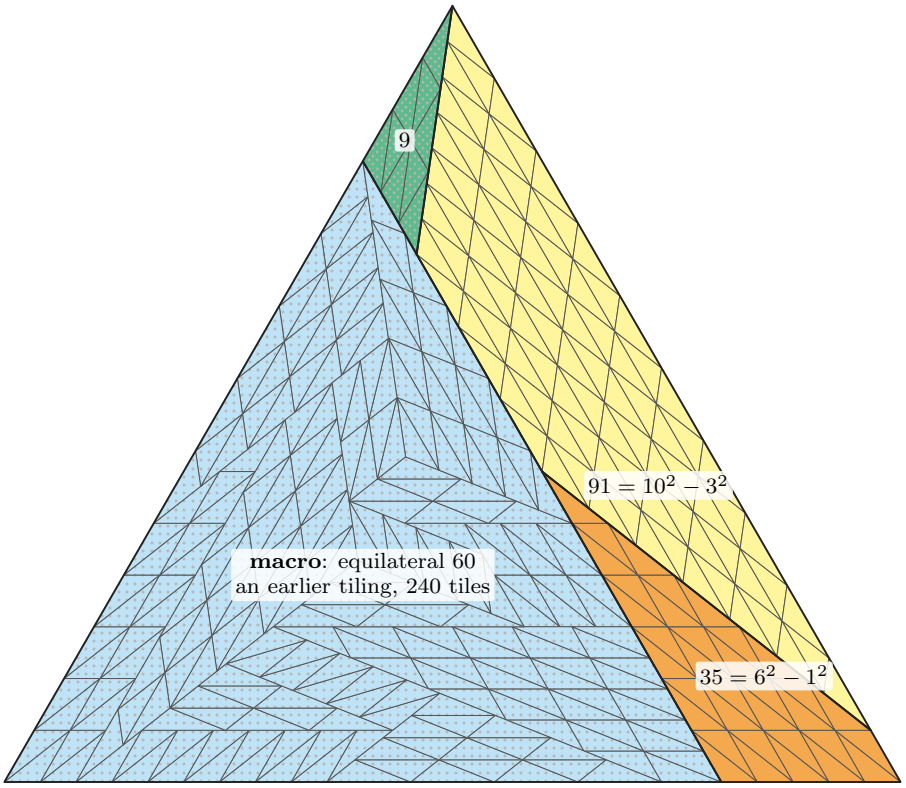


FIGURE 5. Composition by macro pieces, for the equilateral triangle of side 75 tiled by 375 copies of $(3, 5, 7)$: the earlier 240-tile tiling of the equilateral triangle of side 60 (the seed, due to Paliy) is used as one block, and three further pieces of 91, 35 and 9 tiles complete the triangle ($375 = 240 + 91 + 35 + 9$). The same method, with the macro piece a scaled copy of the target of a known tiling, produces the counts of Theorem 5.1 (here the ratio of the macro piece to its known target is $k = 1$).

5.3. **Other counts shipped in this release.** Twelve counts that are squares times earlier template counts (46 500, 104 625, 267 840, 290 625, 418 500, 476 160, 487 060, 569 625, 602 640, 671 460, 744 000, 941 625; module `Draft/TemplateSquares`) are Lean theorems by Proposition 2.2; they were already tile counts mathematically and had been omitted by the census bookkeeping. The thirteen tilings of W_M with $M < f$ of Appendix F are also Lean theorems (`Draft/AllWBelow`), as are their upward closures (`Draft/AllDebtB`); they are subsumed by Theorem W but were proved first. 4830 is a tile count: Paliy's certificate (4830 copies of $(24, 11, 31)$ tile the triangle $(961, 1155, 1426)$) has been converted and verified exactly ([COMPUTED, EXACT]; the construction is Paliy's, [CLAIMED BY PALIY]; his family of mixed-corner layouts gives further values that we have not verified).

6. What is not proved

- **The exclusion half of Conjecture C.** No description of the set of tile counts above 200 is proved; the census has 62 421 undecided non-squarefree values in version 4.0 (63 602 in version 3.0, 63 525 then if the hand-proved Theorem R2 is accepted) and about 770 squarefree values $\le 10^6$.
- **The no side of Theorem W for $e > f/2$:** for each row exactly the scales M with $(M-1)(f-e) < f < (M+1)(f-e)$ (at most two) are undecided, except those excluded by Theorems 7.1 and R2 (for example $W_2(5, 8)$, $N = 412$); and $e = 1$ **and** $e = f - 1$ **for** T_M (for example $T_3(1, 4)$); the T side in general (the band theorem of Section 8 adds $f \le 2Me < f + e$ to $2Me < f$).
- **V and A' for $\gamma = 2\alpha$:** the evidence of Appendix D is not a proof; $(7, 12)$ is incomplete (95.06%).
- **Θ and S_3:** the thresholds of the θ-isosceles family and of the base-α rows; the squarefree cores 465, 966, 2485 ($g = 1$); Lean certificates for the tile counts 1860, 4185, 7440 (exact-verified only here).
- **$15m^2$ for every $m \ge 4$ as a Lean theorem**; the collar step for arbitrary height h (hand only); Paliy's claims that we have not verified (134 values $\le 10^6$ in the census; the mixed-corner family beyond 4830).
- **The squarefree values:** about 770 open; the squarefree tile counts known to us are of the classical forms, or come from a short list of sporadic and layout values (the 22 of the earlier release, 77, 143, 4830).
- **Outside check.** The Lean statements and the proofs have been compared and audited only inside this project, including the statement of `theoremP_6` (the prime case of the earlier releases) and of Theorem W; a check by a reader outside the project has not been made, and for the prime theorem the author makes no claim beyond: stated and machine-checked in the Lean development, published 15 October 2026, internal audits only (Section 8).
- **The band is not the whole of $M < f$:** instances with $d = 2Me - f \ge e$ (19 151 census values) and $e = 1$ (39 values) are untouched by the band theorem; the theorem is Lean with an internal audit only (Section 8).
- **The thick regime beyond rung $e + 2$:** for $M < f$, $Me \ge f$ and $M(f - e) \ge e + 3$ (and $< f$) the question is open; Theorem R2 is a hand proof read only inside the project; see Section 7.
- The search that found the macro layouts is not part of this release and is not described further than in Section 5; its outputs are verified independently.

7. The thick regime: no-tiling rungs

This section is about the scales M below f for which Theorem W does not apply and the exclusion of the earlier release does not either. Write $d = f - e$ and let $2 \le e < f$ be coprime. Theorem W covers $M \ge M^*$; the earlier exclusion (I2 of Paper W) covers $Md \le e$ and $Me < f$. The new statements push the exclusion to $Md \le e + 2$, when $M < f$ and $Me \ge f$.

Theorem 7.1 (rung $e+1$; [PROVED IN LEAN] for the statement, hand proof audited inside the project)**.** *Let* $\gcd(e,f) = 1$, $2 \le e < f$, $M < f$, $Me \ge f$ *and* $M(f-e) \le e+1$. *Then* $W_M(e,f)$ *is not tiled by copies of* $\mathsf{tile}(e,f)$.

Lean: `Draft.SplusThick.wTilable\_not\_of\_small\_M` (module `SplusThickFinal`; aggregate `Erdos634.AllSplusThick`; standard axioms, no `sorry`). The proof studies the side AQ of W_M that ends at the corner 2α: the thick corner-run theorem (Lean `Draft.SplusThick.cornerrun\_counts\_thick`) gives a representation of its length by a, b, c with $z \ge e+1$ tiles having their c-edge on the side, and $z \ge e+2$ when $e+2 \le f$, while the arithmetic of the side (Lean `Draft.SplusThick.C\_AQ'`) gives $z \le M(f-e) \le e+1$; these contradict. The hand proof was audited inside the project (verdict: sound, with fixes now applied: the hypothesis $M < f$, and the arithmetic of AQ for $M(f-e) < f$); nobody outside the project has read it. *Census.* There are 124 non-squarefree values $N = M^2(2f^2 - e^2) \le 10^6$ that are excluded by this theorem and by no earlier one. They are census-level exclusions, except $N = 412 = W_2(5,8)$, which is a Lean theorem about 412 itself: `Draft.NSQM.no\_dissection\_412` (the S+ kill composed into the exclusion chain; modules `NSQRWT2`, `NSQMMainZETW2`, `NSQMMainZETWD2`). The value 270 is a separate Lean exclusion (`Draft.NSQM.no\_dissection\_270`, through the gap lemma for $30s^2$). The statement $W_2(5,8)$ was before this release the example of an open case with $e > f/2$ (Paper W, Section 6): it is now excluded, so the "computed, not proved" search there is superseded.

Theorem 7.2 (Theorem R2, rung $e+2$; [PROVED BY HAND] only)**.** *Let* $\gcd(e,f) = 1$, $2 \le e$, $e+3 \le f$, $M < f$, $Me \ge f$ *and* $M(f-e) \le e+2$. *Then* $W_M(e,f)$ *is not tiled.*

Status. Hand proof, written out in `docs/THEOREM_R2_PROOF.md`. It was audited twice inside the project by readers of the same model family as the author (verdicts: sound with statement-level fixes, applied), re-checked by exact rational computation at twelve pairs (e,f), and tested once by an exact region search at $(e,f,M) = (4,7,2)$, which did not refute it. It has *not* been read by a person outside the project. Lean: only the arithmetic half (`Draft.SplusR2.r2\_arith`) and a corollary that is *conditional* on the geometric core, a statement `R2Count` that is not proved in Lean (`Draft.SplusR2.wTilable\_not\_of\_small\_M2\_of\_R2`). So Theorem R2 is not a Lean theorem, and the numbers that use it are reported in a separate row of the census. *Sketch.* A tiling of W_M has a run C_1^k of tiles with their c-edges on the side AQ at the corner 2α; the earlier work shows $k \ge e+1$ and the case $k = e+1$ (Theorem 7.1). For $k = e+2$ one looks at the tile S_0 in the free wedge at the second lattice point of the run. If it is the mirror image $N_0(0)$ of the first run tile in its a-edge, the chord along its c-edge forces (partition property, $e \ge 2$) a b-edge on the next line, and the row above the run is c^e or a^f; arithmetic shows $c - b$ and $a - b$ are not in the semigroup $\langle a, b, c\rangle$, so the end of a lattice edge must lie inside an edge of a tile, and two tiles then overlap in a wedge of angle β at that point. If S_0 is the half-turn image of the first tile, the row at level 1 is c^{e+1} or $c\,a^f$, and the second is killed by a size-one instance of the corner lemmas. The remaining call has size $e+1$ at level 1 and is killed by the same two ideas. Hypotheses: $e \ge 2$ (the representation of c is unique), $e+3 \le f$ (multipliers stay $\le f-1$). *Census.* For coprime $e < f \le 40$, $M \ge 2$, $Me \ge f$, $M(f-e) = e+2$ there are 26 triples (first: $(4,7,2)$, $N = 328$, and $(7,10,3)$, $N = 1359$); the census program counts 77 more

excluded values $\le 10^6$ with Theorem R2: 199 441 / 129 108 / 63 525. The first rung not covered is $M(f-e) = e+3$: for $(5,9,2)$ and $(7,12,2)$ (for $(6,11,2)$ one has $M(f-e) = 10 = e+4$).

Lemma 7.3 (scale two; [PROVED BY HAND], audited once inside the project)**.** *Let $e > f/2$, $\gcd(e,f) = 1$, $g = f-e$, and suppose $W_2(e,f)$ is tiled. Then the side OB (from the corner 2α to the corner β; length $2fb$) carries exactly $2g$ tiles with an a-edge and $2g$ with a c-edge, and none with a b-edge; the corner run C_1^k has $e+1 \le k \le 2g$.*

The proof is a congruence argument on the representation $2fb = xa + yb + zc$ together with the corner-run theorem of the earlier release (no representation with $z \ge f$ since $2g < f$). With Theorems 7.1 and 7.2: $W_2(e,f)$ has no tiling whenever $2g \le e+2$; what remains is the window $e+3 \le k \le 2(f-e)$, with the smallest cases $(e,f) = (5,9)$ $(k=8)$, $(6,11)$ $(k=9,10)$, $(7,12)$ $(k=10)$. This is *not* proved: it is the open residual of the scale-two case.

Boundary words of the known tilings (computed). For all 37 tilings of $W_M(e,f)$ and 15 of $T_M(e,f)$ among the exact witnesses (all of them layouts of the template of Theorem W) the three sides read, with $n = \lfloor eM/f \rfloor$, $\rho = eM \bmod f$, $\delta = [\rho = 0]$ and $t = n - \delta$: $OA = c^{fM}$, $OB = c^{fM-er}a^{fr-eM}$, $AB = a^{fM-et}c^{ft}b^{eM-ft}$, with $r \in \{n+1, M, M+1\}$ (the three values occurring). That is the vertex table of Theorem W read backwards; the observation that only these three r occur is [COMPUTED, EXACT] on 52 witnesses and is not proved. The b-run on AB is the least residue of eM modulo f, in every witness. Every interior chain of collinear same-side edges is monochromatic or a word $c^p a^q$ or $a^p c^q$ (in 15 678 interior chains), so no Sturmian or Beatty word occurs on lines. *A Beatty form of the threshold* (identity, [COMPUTED, EXACT]: checked exactly for all 296 736 coprime triples with $2 \le e \le f-2$, $f \le 80$, $1 \le M \le 3f$, no mismatch; not proved here): with $n = \lfloor eM/f \rfloor$, "$Me \ge f$ and $(M-1)(f-e) \ge f$" holds if and only if $n \ge 1$ and there is an integer $q \le M-1$ with $e(n+1)/f < q < fn/e$. So the threshold of Theorem W says that this interval has caught an integer; below the threshold, for example $(5,9,2)$, $(6,11,2)$, $(7,12,2)$, it contains none. This restates the threshold and is not a new obstruction.

Dead routes, with data (notes, not proofs). (i) A conjectured lower bound $r \ge 2f - e$ for the corner run ("R^*") is refuted by the scale-four witnesses $W_4(4,7)$, $W_4(5,9)$, $W_4(6,11)$, $W_4(7,12)$ $(r = 8, 11, 14, 13$, against $2f - e = 10, 13, 16, 17)$. (ii) A conjectured bound on interior runs ("S_∞") is refuted by 2 430 interior runs in the witnesses. (iii) Describing the second layer of boundary words by a zigzag automaton gives a finite automaton ($K = f+1$ states) of positive entropy, so it cannot single out the tilings. (iv) Counts of Eisenstein-integer solutions and the windows for $\gamma = 2\alpha$ were measured and gave no exclusion. (v) Reading a W_2 as a scale-two problem reduces it to the window of Lemma 7.3 and no further. **What is not proved:** the exclusion for $M(f-e) \ge e+3$ (and $< f$), the no side of Theorem W for $e > f/2$ in general, T_M below M^*, and any outside verification of the prime theorem (Section 8).

8. What is new in version 4.0: the band theorem and the registry

Labels. *Lean (internal audit)*: a theorem of the library, standard axioms only, no `sorry`; the statement and the proof were audited by fresh sessions of the same model family and read line by line twice; no reader outside the project has checked them. Nothing in this section is announced as established. The Lean files of the band theorem and of everything it imports are in the public repository "Formalizing triangle tiling in Lean" (github.com/ElVec1o/formalizing-triangle-tiling-in-lean, commit `d728130`, version 1.0), module `Erdos634/Draft/LaczFormalMJFinal.lean` and its imports; they are not repeated in this release, which was not rebuilt with them. The reconstruction of the fans-and-chords method follows Laczkovich, Beeson and Beeson–Zhang; the band induction is ours.

The band theorem. Let e, f be coprime with $2 \le e < f$ and $M \ge 2$ with $f \le 2Me < f + e$. Then $T_M(e, f)$ has no dissection into congruent copies of $(ef, f^2 - e^2, f^2)$ (`Draft.LaczFormalMJFinal.band\_theorem\_free`, namespace `Erdos634.LaczJunction.Walk`; aggregate `Draft.AllLaczWalk`). The proof is by the induction $M(j)$ on the base word (no j consecutive c-edges of type cBA before a token with an α/γ end), carried out on a lattice frame at all levels; the arithmetic of the band is checked by brute force for $f < 60$ (42 185 cases, no failure). The special cases $d = 2Me - f \le 1$ and $d \le 2$ are named lemmas used inside the induction, not the extent of the theorem: `no\_tiling\_U`, `no\_tiling\_U2`. *Status:* [PROVED IN LEAN] for the statement as printed, with internal audit only (the label "Lean (internal audit)" used in this section means exactly this: not an extra label). Hypotheses exactly: e, f coprime, $2 \le e < f$, $M \ge 2$ (hence $M < f$), $f \le 2Me < f + e$; dissections may use mirror images and need not be edge to edge; $M = 1$ is not covered. The Lean files are in the public repository ElVec1o/formalizing-triangle-tiling-in-lean (release v1.0, file `Draft/LaczFormalMJFinal.lean`); they are not part of the `AllV3` reproduction build of Appendix H. *Not covered:* the instances with $d = 2Me - f \ge e$ ($M < f$; 19 151 census values), the thin side, $e = 1$ (39 values).

Version stamps. Repository releases of the author: v3.0 (Paper W and the companion), v3.1, v3.2 (the number an earlier release of 12 September 2026 carried at the time), and this release 4.0; "v1.13.1" and "v1.16" in the references are the older numbering of the 15 October release and of the version-2.2 paper; the Lean repository "Formalizing triangle tiling in Lean" is at release v1.0; "census program 1.6.0" is a program version, not a release.

Registry and census. The Lean-formal exclusions are stored one instance at a time (3 204 rows, in `data/registry/ basebeta` e f m); a program (census program, version 1.6.0) declares N excluded when every failing leaf of the exclusion chain is killed by such a row. This closes 918 further non-squarefree values $\le 10^6$ (smallest 1964, then 2636, 2972, 3231, 3788; largest 999 452), and changes nothing below 1000. As before, the assignment "every leaf of N is killed" is made by the program and not by the kernel.

	excluded	undecided
version 3.1 (release v3.1; known 129 117, so 199 364 + 129 117 + 63 593 = 392 074)	199 364	63 593
version 4.0 (this release)	200 282	62 421

The figure 200 282 is counted against the version 3.0/3.1 baseline 199 364 (so +918), without the separate hand-proved R2 row (199 441); we have not checked how many of the 918 coincide with the 77 R2 values. The excluded column is stable under all evidence modes we compared. The column of known tile counts is not printed for version 4.0: the count of known values changed since version 3.1 through sweeps and registry rows that have not yet all been rechecked value by value against exact verification and against Paliy's repository, and none is claimed here. The undecided figure is the one given by the registry of exact-verified tilings; the version 3.1 row is as published.

The prime case. The theorem that no prime $N \equiv 3 \pmod 4$, $N > 3$, is a tile count (`theoremP\_6`) is stated and machine-checked in the author's Lean development and was published in the release of 15 October 2026 (Zenodo 10.5281/zenodo.23400433). It has had internal audits only and no outside check, and the author makes no claim beyond that. Parallel work: D. Paliy (candidate classification, repository note of 29 September 2026 [12]); M. Beeson and G. Roşu, arXiv:2607.23453 [3], v2 (16 October 2026) and v3 (21 October 2026), which cites release v3.2 of the author's repository for the isosceles $2\pi/3$ shape. We say nothing about their proofs, which we have not checked.

Status of the census: program output, internal, not read outside the project.

9. Credits, with dates

"Ours": statement and proof produced in this work. "Formalized": another author's statement and construction, our Lean proof. Dates are of first public record (repository commit dates, arXiv versions) unless marked.

result	status here	whose, and when
Prime counts; the base-β reduction	earlier releases	Paliy, first public proof 2026-09-29 [12]; ours independent; exclusion of $e = 1$ public 2026-09-12 (an earlier Zenodo release, numbered v3.2 at the time); Beeson's withdrawal note for arXiv:2607.19572 (v2, 2026-09-24) [8] credits the author, cited here as public record
Targets W_M, T_M; tilings for $M \geq f$	formalized	Paliy (W/β framework, caps, collars: 2026-10-03; all $M \geq f$ (his v): 2026-10-10); Beeson and Roşu (the targets, the seed)
Theorem W, six-piece template	ours	template found by our search; the setting and the earlier thresholds are Paliy's and Beeson's; the sharpness side ($Me < f$) is from the earlier releases
α up-set, form $b\kappa K^2$, scale additivity	Paliy's; Lean here	Paliy, 2026-09-29 [13]; seeds 84, 189: Beeson (Figs. 28–29), 189: Harries
Collar step, parametric seed $e = 1$, tails	ours	independent of Paliy's construction; credit for the up-set and the threshold K_0 as above
$465g^2$, $g = 2, 3, 4$ (exact tilings)	ours (data)	seeds below Paliy's threshold; exact-verified, Lean planned
Thick regime (no tiling for $M(f-e) \leq e+1$, hand proof of $e+2$)	ours	uses the run framework of Beeson and Roşu and Paliy's W/β setting; the arithmetic of the corner side and the partition property are from earlier releases
Macro pieces	ours (layouts, Lean)	macro-dissections: Laczkovich; Herdt (1125); seeds $15m^2$: Paliy (2026-10-08)
$15m^2$ for all $m \geq 4$, 4830, mixed-corner family	Paliy's ([CLAIMED BY PALIY]); 4830 verified here	Paliy, 2026-10-08
990, 264, 897, 240, 312	earlier releases	Paliy: 990 on 2026-10-06, 264 on 2026-10-07; the others as dated in the earlier releases
θ-isosceles family, scale additivity	Paliy's ([CLAIMED BY PALIY])	Paliy, 2026-09-29
Run induction, Theorem F and D'	earlier releases	method: Beeson and Roşu (preprint of 2026-10-16); $g = 1$: Beeson 2012
1125-tiling	earlier releases	Herdt, as published by Beeson [2]
88	earlier releases	Harries
$2\pi/3$ families F_3, $15m^2, 33m^2$	formalized	Zhang (arXiv:2512.22696); seeds Paliy, Herdt
Rep-tiling and F_0	—	Snover, Waiveris and Williams
Classification of pairs (triangle, tile)	—	Laczkovich
Conjecture C, census, W^{3+} evidence	ours	—

The statements "Paliy claims" in this paper are statements in his public repository [13]; we have read his documents and checked some examples, and we say so where we did. Nothing here is a claim of priority over him for the results listed as his; the dates are given so that the order is visible.

Appendix A. Where the proof of Theorem W is

The proof of Theorem W (coordinates, the six pieces, the sign table, the choice of parameters, the exact accounting of what is Lean, hand and computed) is in Paper W [14], Section 4 (with the worked rows in Section 5) and the separate verification record; the algebra is re-run by `code/check_thmW_algebra.py`. It is not repeated here.

Appendix B. Conjecture C: a candidate classification of the tile counts

The earlier papers give the exclusion side and a growing list of tile counts, but no statement of what the set of tile counts *is*. This section states one, as a conjecture, and says exactly what part of it is proved. **It is not claimed to be true.** It is an organising map: a new tile count is either in a family (no news), or extends a family's range, or is a counterexample to the conjecture.

B.1. The statement.

Recall Proposition 2.2: every family below is understood closed under $n \mapsto k^2 n$.

Conjecture B.1 (Conjecture C; [conjecture]). *A positive integer N is a tile count if and only if $N = k^2 n$ for some integer $k \geq 1$ and some core n in one of the families $F_0, \dots, F_8$ of Table 2, with the parameter ranges stated there.*

Remarks. (i) $84 = 21 \cdot 2^2$ and $189 = 21 \cdot 3^2$ are not sporadic: they are the members $g = 2, 3$ of the row $(1, 2)$ of F_5 (earlier lists had them in F_4; $g = 2$ is below the range $g \geq f + 1 = 3$ proved in Theorem 4.2, so 84 is an earlier-release value); $77 \in F_2$; 183 is *not* a tile count (it is among the excluded values ≤ 200; an earlier draft of this section listed it by a misreading, see Appendix E). (ii) Theorem 3.1 is stated for $2 \leq e \leq f - 2$; F_1 in Table 2 includes the two edge members with their threshold f. (iii) For F_6 the earlier papers state two conjectures about its exclusion side, V ($v \mid g$ is necessary) and A' (a size bound); Appendix D gives evidence for them.

B.2. What is proved about C.

Theorem B.2 (The existence half, on the proved ranges; [proved in Lean]). *Let* ConjCBase(N) *be the statement "$N = k^2 n$, $k \geq 1$, n in the Lean predicate $F_0 \cup \cdots \cup F_6$ with the ranges proved at the time of that module". Then* ConjCBase(N) *implies that N is a tile count.*

Lean: `ConjC.Exist.conjC_existence` (modules `ConjC/Defs`, `ConjC/Exist`, `ConjC/BetaCount`; aggregate `AllConjC`; each family predicate is exactly the parameter range of an existing Lean existence theorem, and the non-vacuity of each is checked on an instance, e.g. `conjCBase_143`, `conjCBase_11625`). It collects constructions under one statement; it is not new mathematics beyond the family theorems it cites. The predicate in Lean is narrower than Table 2: it has F_1 for $M \geq f$ only (Theorem 3.1, the tails and the macro counts of Sections 4–5 are separate Lean theorems, not yet folded into it), it has no F_7, F_8, and its F_5 is the part proved at the time (the row $e = 1$, $g \geq f + 1$; $966g^2$, $g \geq 25$; five engine values; the better tail $g \geq 17$ of Theorem 4.3(c) is a separate theorem not folded into this predicate).

[computed, exact] *Consistency checks* (exact computations in the working notes, repeated for the census of this release where stated). (a) The families generate exactly the 105 tile counts ≤ 200 of the earlier theorem, no more and no fewer. (b) None of the 199 364 (version 3.0) Lean-formal excluded values of the census is generated by any family (rechecked for the Theorem W

	cores n	tile, target	range proved (label) / credit
F_0	1, x^2+y^2 $(x,y\geq 1)$, 3, 6	similar tiles, rep-tiles	all [PROVED IN LEAN] / Snover–Waiveris–Williams, Laczkovich
F_1	$2f^2-e^2$ and $3f^2-e^2$, at scale M: $M^2(2f^2-e^2)$, $M^2(3f^2-e^2)$	$\mathsf{tile}(e,f)$; W_M, T_M	$e=1$ or $e=f-1$: $M\geq f$. $2\leq e\leq f-2$: $M\geq M^*(e,f)$ (Thm 3.1). [PROVED IN LEAN] / Paliy, Beeson ($M\geq f$); ours (M^*)
F_2	$(2q^2-p^2)(3q^2-p^2)$, $0<p<q$	five-piece layouts	all [PROVED IN LEAN] / Paliy
F_3	$(2a+b)(a+b)$, $3(a+b)(a+2b)$, $(a+2b)(2a+b)$ and Zhang's five forms $\times m^2$; $15m^2$, $33m^2$	tiles with an angle $2\pi/3$, $c^2=a^2+ab+b^2$; $(3,5,7)$	ranges as in the Lean predicates; $15m^2$: $m\in\langle 4,9\rangle$, and $m=5,6,7$ individually [PROVED IN LEAN] / Paliy, Zhang, Herdt. $15m^2$ for all $m\geq 4$: [CLAIMED BY PALIY]
F_4	88, 143 (sporadic)	tile $(3,5,7)$ on the triangles $(21,55,56)$ (88) and $(39,49,55)$ (143)	[PROVED IN LEAN] / Harries (88); ours (143)
F_5	$(f^2-e^2)(2f^2-e^2)\,g^2$, $g\in G(e,f)$	$\mathsf{tile}(e,f)$; $B_g(e,f)$	$e=1$: $g\geq f+1$; $(1,4)$: $g\geq 5$; $(1,6),(2,5)$: $g\geq 2$; $(2,3)$: $g=3,4,5$ and more by Thm 4.1 [PROVED IN LEAN] / [PROVED BY HAND]; Paliy: eventual up-set in every row, [CLAIMED BY PALIY]
F_6	$b\,(v(D+\tau v))^2$, positive integers D,τ (four more integers are subject to two linear Diophantine conditions, (E1), (E2) of the version-2.2 paper); 1125, 19 600, 130 977	$\gamma=2\alpha$, tile (u^2,v^2-u^2,uv)	Herdt's family and its uniform version [PROVED IN LEAN] / Herdt, Laczkovich
F_7	4830	$(24,11,31)$	[COMPUTED, EXACT] / Paliy, [CLAIMED BY PALIY] (mixed-corner family)
F_8	bt^2, $t\in S(e,f)$	$\mathsf{tile}(e,f)$; θ-isosceles target (legs tfb, base teb, base angles $\alpha+\beta$, bt^2 tiles)	[CLAIMED BY PALIY] / Paliy (semigroup $S(e,f)$, $te\geq 2f$ necessary [PROVED IN LEAN])

TABLE 2. The families of Conjecture C and what is proved about each. $\langle 4,9\rangle$ is the set of non-negative combinations of 4 and 9. In F_1 the ranges are those of Theorem 3.1 and of the earlier theorem for $M\geq f$; for the members $e=1$ and $e=f-1$ the threshold f is proved sharp for W_M.

family and the macro counts in this release: no overlap). (c) Every one of the known tile counts of the census is generated by some family; this check is partly circular, because the known values

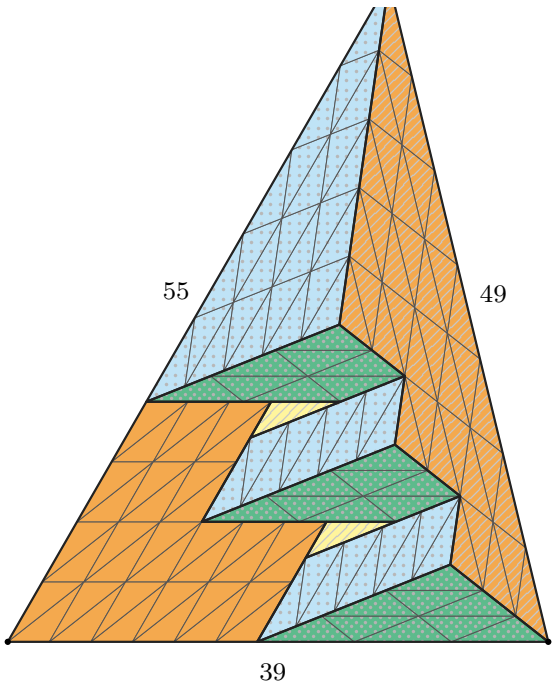


FIGURE 6. The tiling of the triangle $(39, 49, 55)$ by 143 copies of $(3, 5, 7)$ (angle $2\pi/3$ between the sides 3 and 5), the sporadic count 143 of F_4; it was found by a search in an earlier release and is a Lean theorem there. 143 is not of any of the forms k^2, $x^2 + y^2$, $3k^2$, $6k^2$ and is in no parametric family of this paper. Colours are orientation classes (same colour: tiles related by translation or half-turn); hatched: mirror images.

were found by these families, so it shows that the families are complete for what is known and nothing more. The sharp content of C is the exclusion half.

B.3. **What the exclusion half needs.** Everything proved in the exclusion direction in the earlier releases is compatible with C. The exclusion half of C is not proved: it is the statement that no value outside the families is a tile count, and it decomposes into threshold statements, one for each family at the place where the known range stops:

- W^* for F_1 with $2 \le e \le f - 2$: $W_M(e, f)$ is tiled iff $M \ge M^*(e, f)$; open for $e > f/2$ (Section 3); and T^*: T_M is tiled iff W_M is.
- V, A' for F_6: the target Ω_g of $\gamma = 2\alpha$ is tiled only if $v \mid g$, and for $g = 1$, $(g - 1)(v - u) < u$; open.
- Θ for F_8: $\theta_t(e, f)$ is tiled iff t is in a semigroup, with $te < 2f$ excluded (proved, [PROVED IN LEAN]).
- S_3 for F_5: the row $G(e, f)$ has a smallest element $g_{\min}(e, f)$ and its gaps; known values are $g_{\min}(1, 2) = 2$ (exactly, since 21 is not a tile count: every $N \le 200$ is decided in the earlier release), $g_{\min}(2, 5)$, $g_{\min}(1, 6) \in \{1, 2\}$, $g_{\min}(1, 4) \le 5$.

Z for F_3: Zhang's counts m^2ab etc. below the threshold $m < 3t_0$ of the earlier release are not tile counts unless another family covers them; the known exceptions are the $15m^2$ and $33m^2$ counts.

- small leaves: the $\pi/3$ and $2\pi/3$ pairs with irrational angle (closed, [PROVED IN LEAN]), the commensurable cases.

Each of these is an infinite non-existence theorem; the methods of this project (additive invariants and the run induction of the earlier papers) prove them only in the ranges stated, and a count at a single scale cannot decide a threshold statement at all, because the same tile tiles the target at every larger scale. **Expectation.** We expect the true set to be a union of schemas of this type, since a threshold or a semigroup appears in all constructions known; whether the thresholds $g_{\min}(e,f)$ and $M^*(e,f)$ have closed forms for all rows is open (for F_1, W now has one). If a core outside all families is ever found, C needs a further family and the classification becomes an open-ended census. **Falsifiability.** The cheapest tests of C: the smallest value of F_5 outside the known range ($1860 = 465 \cdot 2^2$, 4185, 7440), the squarefree cores 465, 966, 2485 ($g = 1$; C predicts that they are not tile counts), and the first open value of F_6 at $g = v$, namely $1617 = (4, 7, g = 7)$.

Appendix C. The census

C.1. **What it counts.** The universe is the 392 074 non-squarefree integers $N \in [2, 10^6]$; N is *excluded* if the decidable arithmetic hypothesis of the single Lean theorem of the earlier paper (the exclusion theorem; it is not restated here) fails for it (a program evaluates the hypothesis for each N; the theorem is Lean, the evaluation is computation), *known* if some construction gives a tiling, and *undecided* otherwise. The three sets are disjoint (asserted by the script: no known value is excluded). The baseline is the census of the previous release.

	excluded	known (tile counts)	undecided
version 2.2 / release v1.16	199 140	102 613	90 321
version 3.0 / release v3.0 (Lean-formal exclusions)	199 364	129 108	63 602
change	+224	+26 495	−26 719
version 3.0 with the hand-proved class R2 (separate row)	199 441	129 108	63 525
version 4.0 (Section 8; known column not printed)	200 282	–	62 421

The exclusion side grew by 224 values: 124 by the thick-regime theorem of Section 7, 91 by two further classes (85 values of the form $30s^2$ through the earlier gap lemma and 6 values $t^2(12t^2-1)$ with $e = 1$, $2M = f$), and 9 from the re-run of the census program; the exclusion assignment is made by the census program, and only 412 and 270 below 1000 are also Lean theorems about *that* N (`no_dissection_412`, `no_dissection_270`); the others are census-level (their leaf checks are too heavy for the kernel in the current interface). The row with Theorem R2 adds 77 values and is *hand-proved*; do not read it as formal. The 26 495 new known values, by label:

label	values	source
[PROVED IN LEAN] (formal)	26 312	Theorem 3.1 family (26 138 new), collar and tails, macro counts, 13 W-below closures, k^2-multiples
[COMPUTED, EXACT] only (exact tiling, no Lean)	145	witness-only values from template searches
[PROVED BY HAND] (hand)	35	collar step with $h > 1$ on the rows $(1,f), (2,3), (2,5)$ (Theorem 4.1)
paper-formal closure over exact seeds	3	upward closure statement of the earlier paper over verified seeds

(The 102 613 known values of the baseline keep the labels of the previous release.) Theorem W alone moves 26 138 values from undecided to known: the family has 44 495 values $\le 10^6$ (28 946 from W, 18 431 from T, union 44 495; the largest f is 251); 18 357 of them were already known by other means, none lies in the excluded set. **Paliy-claim-only values.** A further 134 values $\le 10^6$ lie in families that Paliy claims ([CLAIMED BY PALIY]; examples are 1380, 2024, 3105, 4554, 5520, 8096, and the values $15m^2$, $m \ge 4$, that no Lean theorem covers, such as $1815 = 15 \cdot 11^2$) and that we have neither proved nor verified; they are *not* counted as known, they are among the 63 602 undecided. The three values 1860, 4185, 7440 ($465g^2$, $g = 2, 3, 4$) were found after this census was run; they are exact-verified tile counts and are not in the table.

C.2. **Squarefree values.** Of the squarefree $N \le 10^6$, about 770 remain open at the exact level (771 in version 2.2: the difference is 4830, a tile count by Paliy's certificate, exact-verified here); at the Lean level the count of the previous release (791) is unchanged, since all new Lean theorems concern non-squarefree values. The open squarefree values ≤ 1000 are 231, 329, 354, 385, 455, 465, 517, 777, 806, 935, 966. The undecided non-squarefree values in $[201, 1000]$ were 38 in the census of the first draft of this release; of these, 270 and 412 are now excluded by Lean theorems and 750 and 963 by the census program only, leaving 34: 236, 280, 284, 297, 332, 351, 420, 423, 444, 480, 524, 531, 552, 594, 616, 644, 668, 716, 732, 747, 816, 824, 840, 855, 876, 880, 908, 912, 924, 927, 944, 945, 956, 999 (414 and 639, tile counts of $W_3(2, 5)$ and $T_3(2, 5)$, are no longer on the list). Every $N \le 200$ was already decided.

C.3. **What the census is not.** It is not a classification: the undecided values may be tile counts or not. It rests on the families known to us; a tile count in a family we do not know is invisible to it. The excluded values are excluded by Lean theorems, but the assignment "this N satisfies the hypothesis" is made by a program, not by the kernel (as in the earlier paper). The known values are of the stated labels; only the [PROVED IN LEAN] ones are Lean theorems about *that* N.

APPENDIX D. COMPUTER EVIDENCE ON ONE OPEN FAMILY: THE W^{3+} CENSUS

This section records exact computations that are *not* proofs and *not* formal.

D.1. **The family and the question.** Let $u < v < 2u$ be coprime, $b = v^2 - u^2$, tile (u^2, b, uv) (this is the family F_6, with $\gamma = 2\alpha$), and Ω_g the isosceles target with base angles α, legs gub, base gvb; a tiling has $g^2 b$ tiles. The earlier papers prove that Ω_g is not tiled when $(g-1)(v-u) < u$ (Theorem F) and conjecture that it is tiled only if $v \mid g$ (V) and $g \ge g_D(u, v)$ (A'); the only tilings known are three Herdt-type counts, 1125, 19 600, 130 977. Two open instances with $v \nmid g$ and $(g-1)(v-u) \ge u$ are $g = 3$ for $(u, v) = (4, 7)$ (tile $(16, 33, 28)$, $N = 297$) and $(5, 8)$ (tile $(25, 39, 40)$, $N = 351$).

D.2. **The words.** In a tiling of Ω_g, the tiles with an edge on a leg partition the leg into segments; reading from the corner α to the apex gives a *word*: the sequence of edge letters (a, b or c) with the angles of the tiles at their ends. The length of the leg and the angle sums at the ends leave a finite list of admissible words; hand-proved filters (named in the working notes; they are not the conjecture A' above) and a rule for the junction at the apex cut it down to the set we call W^{3+}. For each word an exhaustive exact search (integer arithmetic, no floating point) asks whether the part of a tiling near that leg exists. The verdict "no tiling" for a word is therefore conditional on the filters; it covers every tiling whose leg word is that word, and *only* such tilings (it is not a verdict on Ω_g unless it is returned for every word, including the words outside W^{3+} that the filters exclude).

(u, v), $g = 3$	words	no tiling	tiling found	open
$(4, 7)$	191 940 (complete list)	191 940	0	0
$(5, 8)$	84 462 (complete list)	84 462	0	0
$(7, 12)$	8 431 936 799	8 014 982 151 (95.06%)	0	416 954 648

For $(4, 7)$ and $(5, 8)$: all but 323 and 630 words fall within 2 s each; the remainder were rerun at 120 s, 1800 s, and without a time limit (the two hardest words of $(4, 7)$ took 5614 s and 3269 s); every word is decided. For $(7, 12)$ the word list has 8.4×10^9 entries and was covered by an adaptive prefix tree (a "no tiling" for a prefix refutes all its completions: the justification is that no tile edge straddles the end of a word prefix); the residual 4.2×10^8 words are open and the run was stopped because the fraction of prefixes that time out grew with depth. **Controls.** The search finds the known tilings when it is given their own words (the words of the tilings of the scaled tile at $m = 3, 4$, for all three pairs: found and verified), and refutes the reversed words and words refuted independently by a slower program. **What this shows and does not show.** No tiling was found, so V and A' are not refuted. It does not show that 297 and 351 are not tile counts, because the verdict is for the one-leg word set, not for all tilings and all shapes of target; and without a Lean certificate it is not a theorem. It is evidence, and it narrows where a counterexample could be.

Appendix E. Corrections of earlier statements

Statements made in our working notes, and in one case in an earlier release, that this paper corrects. None of the corrections affects a Lean theorem, a prime-case statement or an excluded value of the census.

(1) **183 is not a tile count.** An earlier draft of the conjecture listed 183 as a sporadic tile count (attributed to Laczkovich). 183 is excluded, and every $N \leq 200$ is decided in Lean; the draft misread a statement about $5 \cdot 183^2$. The Lean predicate F_4 never contained it.

(2) **Tiles of T_M and W_M below the scale f exist for $e \geq 2$.** Our notes asserted that $T_M(e, f)$ and $W_M(e, f)$ have no tiling for $M < f$ when $e \geq 2$ ("B_1"), and that the T half of this statement stood after the W half failed. Both are false: Theorem 3.1 and the glue of Paper W, Section 3.3. The version 1 paper [15] recorded, as an observation, that "in every case decided so far" the scale f is the threshold; Corollary 3.3 is a counterexample ($W_4(3, 5)$, $M = 4 < 5$). The open problems of that paper on the scales $f/(2e) < M < f$ are partly closed here (Section 3).

(3) **The template "exists iff $4f < 9e$" is false.** An intermediate note (13 witnesses below f, then a replay of one search) stated that the six-piece template exists only for $4f < 9e$, was absent for 432 coprime pairs with $f \leq 60$, and that the conjectured threshold held only there. This was an artefact: the replay of one search history has an empty cone for the other pairs, but the geometry of the same six pieces works for all $2 \leq e \leq f - 2$ (Theorem 3.1; for $(2, 5)$, $(3, 7)$, $(4, 9)$, $(5, 12)$, $(3, 8)$ the tilings exist).

(4) **The size conjecture is false.** The working notes (and a conjecture in a longer draft) stated that a target not similar to the tile has longest side at least $c^2 = f^4$. $W_4(3, 5)$ has longest side $500 < 625$.

(5) **Paliy's 4830 is a tiling.** Our notes said that his 4830-tile file is not a tiling. We had checked the wrong file (his refuted moment counterexample, not his certificate). His certificate, 4830 copies of $(24, 11, 31)$ on the triangle $(961, 1155, 1426)$, is verified exactly (Section 5).

(6) **The θ-isosceles statement.** Our notes stated "targets $\theta_t(e, f)$ with $te \geq f$ have no tiling". That is false: $\theta_4(1, 2)$ and $\theta_8(2, 3)$ are tiled. The proved statement is the necessary

bound $te \geq 2f$; the true statement is a threshold (a semigroup of scales, Paliy), not an exclusion.

(7) **Priority of the α up-set.** The up-set in g of the base-α rows and the form $b\kappa K^2$ are Paliy's, dated 29 September 2026 (Section 4); our notes had presented the up-set as ours. The scale additivity of the rows is also his.

(8) **The collar corollary has two conditions.** The first version of the statement "g_0 in the row implies every $g \geq g_0$ is in the row" in our notes used only $h_{\max}(g_0) \geq g_0 - 1$; the audit added $h_{\max}(x) \geq 1$ for $x \geq 2g_0 - 1$.

(9) **Counts in a note.** The census effect of the collar step was first stated as 235 values; the correct figure, by an audit that recounted, was 194 (119 in the rows $(1, f), (2, 5)$ and 75 in $(2, 3)$); the figures of this release supersede both.

(10) **The threshold formula $(M-1)(f-e) \geq f$.** Used as a threshold for all rows it contradicts the proved side theorem W_M none if $Me < f$ exactly when $\lceil f/e \rceil > \lceil f/(f-e) \rceil + 1$ (for instance $(2, 7)$: the formula gives $M = 3$, but $3 \cdot 2 < 7$). The correct threshold is $M^*(e, f) = \max(\lceil f/e \rceil, \lceil f/(f-e) \rceil + 1)$.

(11) **The pattern $g_{\min} = f + 1$.** Our notes observed that the first realised scale of a base-α row was $f + 1$ for even f and recorded this as a pattern; $g_{\min}(1, 6) \leq 2$ (Theorem 4.3) refutes it ($f = 6$ is even), and $g_{\min}(2, 5) \leq 2$ as well.

The register of corrections of the project is in the repository (file `TIMELINE.md` records the dates).

Appendix F. Data: the explicit tilings

F.1. **The thirteen first witnesses with $M < f$.** Found first by a computer search, and now instances of Theorem 3.1; every one is a Lean theorem (module `Draft/DebtBWTilable`, with the closure upward in M).

(e, f, M)	W_M tiles	T_M tiles	(e, f, M)	W_M tiles	T_M tiles
$(3, 5, 4)$	656	1056	$(7, 12, 4)$	3824	6128
$(4, 7, 4)$	1312	2096	$(7, 10, 5)$	3775	6275
$(5, 8, 4)$	1648	2672	$(7, 9, 6)$	4068	6984
$(5, 7, 5)$	1825	3050	$(8, 11, 5)$	4450	7475
$(5, 11, 3)$	1953	3042	$(9, 11, 7)$	7889	13 818
$(5, 9, 4)$	2192	3488	$(6, 11, 4)$	3296	5232
$(7, 11, 4)$	3088	5024			

F.2. **The shipped files.** Directory `data/witnesses/v3_exact`: 97 files, each named by the number of tiles, the kind and the parameters (for example `tiling_414_W_2_5_M3.exact.txt.gz`). Each begins with a comment line stating the number of tiles and the side lengths of the tile; every other line gives the three vertices of one tile in exact rational coordinates $(X, Y\sqrt{D})$. Kinds: `W`, `T` (Theorem W and its glue: 37 and 15 files), `collar` (6) and `macro` (38) (Sections 4, 5), `F3_paliy` (Paliy's 4830). Verification: the exact verifier of the repository (`bte verify`) reports, for each file, that the tiles are congruent to the stated tile, that their interiors are pairwise disjoint, that the total area equals the area of the hull, and that the hull is a triangle. All 97 pass.

Appendix G. Index of the Lean statements

All in the library of the repository (directory `lean/`); aggregate `Erdos634.AllV3` (the modules new in this release; build one module at a time through `code/lakeserial.py`); the axioms of 98 of them are printed by `lean/axioms/AxiomsAllV3.lean` and every line reads `[propext, Classical.choice, Quot.sound]`.

statement	Lean name (namespace `Erdos634....`)
Theorem 3.1	`Draft.ThmW.thmW`; counts `Draft.ThmW.tileCount_W`, `tileCount_T`; examples `inst_414`, `inst_656`, `inst_639`, `inst_27324`
13 first witnesses, closures	`Draft.DebtB.closure_656` etc.; `DebtB.counts_of_wTilable`
Collar layer step	`Draft.CollarLayer.layer_core`, `Draft.CollarFam.layer_step`
Theorem 4.2	`Draft.CollarUp.tileCount_e1_up`; instances `tileCount_189`, `tileCount_16740`, `tileCount_159040`
Theorem 4.3	`Draft.CollarFam.tileCount_465_sq`, `Draft.CollarFam5.tileCount_2485_sq5`, `Draft.CollarFam17.tileCount_966_sq17`
Macro pieces (Thm 5.1)	`Draft.MacroPiece.macroTFam_gen`; `Draft.TM375.tileCount_375` etc.; `Draft.T62125.tileCount_62125`; `Draft.AllMacro.tileCount_78246`
Twelve square multiples	`Draft.TemplateSquares.tileCount_46500` etc.
Conjecture C, existence half	`ConjC.Exist.conjC_existence`; `ConjC.BetaCount.tileCount_beta_ge`
Band theorem (Section 8)	`Draft.LaczFormalMJFinal.band_theorem_free`; `Draft.LaczFormalUDims.no_tiling_U`; `Draft.LaczFormalW2Walk.no_tiling_U2`; aggregate `Draft.AllLaczWalk`

Appendix H. Reproducing the checks

From a clean clone, with the toolchain of `lean/lean-toolchain`: build the aggregate one module at a time, print the axioms, and verify the exact tilings:

```
python3 code/lakeserial.py lean build Erdos634.AllV3
(cd lean && lake env lean axioms/AxiomsAllV3.lean) \
   | python3 code/axiom_gate.py
BTE=rust/bonfioli_tiling_engine/target/release/bte
for f in data/witnesses/v3_exact/*.gz; do
  $BTE verify "$f" | grep -q VERIFIED || echo FAIL "$f"
done
```

The build needs about 6 GB of memory for its largest module and about 22 GB of disk for the build products; one module is built at a time. The scripts `code/reproduce.sh` (all earlier checks) and the above give the complete set.

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